

Python for Computer Science and Data Science 1 (CSE 3651)

MAJOR ASSIGNMENT-1: VARIABLES, OPERATORS, CONTROL STRUCTURES, AND FUNCTIONS

Assignment Questions

1. Write a Python function basic salary that accepts two parameters: hourly rate and hours worked per week. The function should calculate the basic salary per month (assuming a month has 4 weeks). If the hours worked per week exceed 40, create another function overtime salary, where every extra hour worked is paid at 1.5 times the normal hourly rate. Finally, create another function total salary that returns the sum of the basic salary and overtime.

```
In [1]: def basicSalary(hourlyRate, hoursWorkedPerWeek):
        regularHours=hoursWorkedPerWeek
        if hoursWorkedPerWeek>40:
            regularHours=40
        monthlyBasic=hourlyRate*regularHours*4
        return monthlyBasic

def overtimeSalary(hourlyRate, hoursWorkedPerWeek):
    overtimeHours=0
    if hoursWorkedPerWeek>40:
        overtimeHours=hoursWorkedPerWeek-40
    overtimeRate=hourlyRate*1.5
    monthlyOvertime=overtimeHours*overtimeRate*4
    return monthlyOvertime

def totalSalary(hourlyRate, hoursWorkedPerWeek):
    basic=basicSalary(hourlyRate, hoursWorkedPerWeek)
    overtime=overtimeSalary(hourlyRate, hoursWorkedPerWeek)
    return basic+overtime

rate=float(input("Enter Hourly Rate: "))
hours=float(input("Enter Hours Worked/Week: "))
print("The Rate And Hours Entered Are: ", rate, " ", hours)
salary=totalSalary(rate, hours)
print(f"Your Estimated Total Monthly Salary Is: {salary}")
```

```
The Rate And Hours Entered Are: 300.0 500.0
Your Estimated Total Monthly Salary Is: 876000.0
```

2. Create a function tax amount that shows how much taxes are deducted from the basic salary. Taxes are applied as follows:
- If the salary is less than Rs. 60,000/-, deduct 10% as tax.
 - If the salary is between Rs. 60,000/- and Rs. 85,000/-, deduct 15% as tax.
 - If the salary is more than Rs. 85,000/-, deduct 20% as tax.

```
In [2]: def calculateTax(basicSalary):
        tax=0
        if basicSalary<60000:
            tax=basicSalary*0.10
        elif basicSalary<=85000:
            tax=basicSalary*0.15
        else:
            tax=basicSalary*0.20
        return tax

salaryInput=float(input("Please Enter Your Basic Salary: "))
taxAmount=calculateTax(salaryInput)
```

```
print(f"Your Basic Salary Is: {salaryInput}")
print(f"The Tax You Need To Pay Is: {taxAmount}")
```

Your Basic Salary Is: 500000.0
The Tax You Need To Pay Is: 100000.0

3. Using the function basic salary from Question 1, write another function gross salary that calculates the gross salary of an employee. This function should accept basic salary as input (output from Question 1), consider a fixed value of allowances (e.g., 20% of basic salary), and return the gross salary (basic salary + allowances - tax).

```
In [3]: def calculateTax(salary):
        tax=0
        if salary<60000:
            tax=salary*0.10
        elif salary<=85000:
            tax=salary*0.15
        else:
            tax=salary*0.20
        return tax

def calculateGross(salary):
    bonus=salary*0.20
    tax=calculateTax(salary)
    totalPay=salary+bonus-tax
    return totalPay

basicSalary=float(input("What Is Your Basic Salary? "))
finalPay=calculateGross(basicSalary)
print("Your Entered Basic Salary Is: ",basicSalary)
print(f"Your Gross Salary Will Be: {finalPay}")
```

Your Entered Basic Salary Is: 700000.0
Your Gross Salary Will Be: 700000.0

4. Using the gross salary function from Question 3, write a function salary bracket that categorizes the employee's gross salary into one of the following brackets:

- "Low income" if gross salary is below Rs. 50,000/-.
- "Middle income" if gross salary is between Rs. 50,000/- and Rs. 80,000/-.
- "High income" if gross salary is more than Rs. 80,000/-.

```
In [4]: def getSalaryBracket(grossSalaryInput):
        if grossSalaryInput<50000:
            return "Low Income"
        elif grossSalaryInput<=80000:
            return "Middle Income"
        else:
            return "High Income"

userSalary=float(input("Enter Your Gross Salary:"))
print("Your Entered Gross Salary Is: ",userSalary)
incomeLevel=getSalaryBracket(userSalary)
print("Your Salary Bracket Is: ",incomeLevel)
```

Your Entered Gross Salary Is: 300000.0
Your Salary Bracket Is: High Income

5. Take three different sets of employee names, hourly rates and hours worked per week as user input. Write a Python function employee report that generates a formatted report of all employees' salary details. This function should print the employee names, basic salaries, gross salaries, tax amounts, and salary brackets in a readable format.

```

In [5]: def basicSalary(hourlyRate, hoursWorkedPerWeek):
    regularHours=hoursWorkedPerWeek
    if hoursWorkedPerWeek>40:
        regularHours=40
    monthlyBasic=hourlyRate*regularHours*4
    return monthlyBasic

def taxAmount(basicSalaryInput):
    tax=0
    if basicSalaryInput<60000:
        tax = basicSalaryInput*0.10
    elif basicSalaryInput<=85000:
        tax=basicSalaryInput*0.15
    else:
        tax=basicSalaryInput*0.20
    return tax

def grossSalary(basicSalaryInput):
    allowances=basicSalaryInput*0.20
    tax=taxAmount(basicSalaryInput)
    gross=basicSalaryInput+allowances-tax
    return gross

def salaryBracket(grossSalaryInput):
    if grossSalaryInput<50000:
        return "Low Income"
    elif grossSalaryInput<=80000:
        return "Middle Income"
    else:
        return "High Income"

def employeeReport():
    employeeData=[]
    print("Three Employee Details:")
    for i in range(3):
        print(f"\nEmployee {i + 1}:")
        name=input("Enter Name: ")
        print("Entered Name: ",name)
        rate=float(input("Enter Hourly Rate: "))
        print("Entered Rate: ",rate)
        hours=float(input("Enter hours worked/week: "))
        print("Entered Hours: ",hours)
        print()
        employeeData.append([name,rate,hours])
    print("\n"+"-"*80)
    print("Employee Salary Report".center(80))
    print("-"*80)
    header=f"{'Name':<20} {'Basic Salary':<15} {'Tax Amount':<15} {'Gross Salary':<15} {'Bracket':<15}"
    print(header)
    print("-"*80)
    for emp in employeeData:
        name=emp[0]
        rate=emp[1]
        hours=emp[2]
        basic=basicSalary(rate, hours)
        tax=taxAmount(basic)
        gross=grossSalary(basic)
        bracket=salaryBracket(gross)
        reportLine=f"{'name':<20} {'basic':<15.2f} {'tax':<15.2f} {'gross':<15.2f} {'bracket':<15}"
        print(reportLine)
    print("-"*80)
employeeReport()

```

Three Employee Details:

Employee 1:
Entered Name: Amulya
Entered Name: Amulya
Entered Rate: 400.0
Entered Hours: 500.0

Employee 2:
Entered Name: Deep
Entered Rate: 600.0
Entered Hours: 700.0

Employee 3:
Entered Name: Shrivastava
Entered Rate: 600.0
Entered Hours: 900.0

Employee Salary Report				
Name	Basic Salary	Tax Amount	Gross Salary	Bracket
Amulya	64000.00	9600.00	67200.00	Middle Income
Deep	96000.00	19200.00	96000.00	High Income
Shrivastava	96000.00	19200.00	96000.00	High Income

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